

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 2432

II

As specified in the official PDF

Source-grounded

Fundamentals of Computer Programming

Study the official course objectives in the source syllabus.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 2432 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 2432.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma Programme
Course code	2432
Course title	Fundamentals of Computer Programming
Semester	II
Course category	As specified in the official PDF
Periods per week	As specified in the official PDF
Periods per semester	As specified in the official PDF
Credits	As specified in the official PDF
Prerequisite	Topic Course code Course Title Semester Basic computer knowledge Logical reasoning 1008 Foundational IT Skills I Course Outcome CO Course Outcome Blooms Taxonomy Level CO1 Understand the fundamentals of C programming concepts and control structures. Understand CO2 Apply arrays, strings, structure and union Apply Develop modular programs using functions, recursion, searching and sorting techniques to solve CO3 Apply problems. CO4 Use pointers and file handling techniques effectively. Apply CO-PO mapping COURSE ARTICULATION MATRIX PROGRAM OUTCOMES
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

47 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES**Objectives, outcomes and mapping****Course objectives**

- Study the official course objectives in the source syllabus.

Course outcomes

CO	Official outcome	Study level
CO1	3 - 1 - - - - - - -	See official syllabus
CO2	3 - - - - - - -	See official syllabus
CO3	3 - - - - - - -	See official syllabus
CO4	3 - - - - - - - Total 12 - 1 - - - - - - Correlation Level 3 - 1 - - - - - - Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-" Teaching and Assessment Scheme Examination Scheme Teaching Scheme/Week Self-learning Total Credits Theory Practical (SS)/Week Hours/Semester C Total L T P S CIE ESE Total CIE ESE Total 3 1 0 0 5.5 60 4 40 60 100 0 0 0 100 L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination Diploma Curriculum Revision 2026 2432 Page #1 Syllabus - Major Topics Instructional Hours Module Title Major Topics L T • Introduction to programming paradigms • Applications of C language • Structure of a C program • Character set, constants, identifiers, keywords • Data types, variables • Expressions and statements	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 - 1 - - - - -
CO2	CO2 3 - - - - -
CO3	CO3 3 - - - - -
CO4	CO4 3 - - - - -

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	C Fundamentals 11 3	12	As specified
Module 2	Arrays & Strings • Two-dimensional arrays 11 4	13	As specified
Module 3	Searching and Sorting 12 4	13	As specified
Module 4	Function, Pointers & File 11 4	9	As specified

MODULE 1**C Fundamentals 11 3**

This module is presented from the official syllabus scope for Fundamentals of Computer Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: C Fundamentals 11 3
- Official duration: As specified
- Official topic entries captured: 12
- The full source excerpt is preserved below for coverage checking.

▼ C Fundamentals 11 3

C Fundamentals 11 3 is an official topic in Module 1 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify C Fundamentals 11 3 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, c fundamentals 11 3 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What C Fundamentals 11 3 means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- C Fundamentals 11 3 definition/meaning . steps, application . Technical terms English- .

▼ • Input and output functions

• Input and output functions is an official topic in Module 1 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Input and output functions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • input and output functions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Input and output functions means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഇൻപുട്ട് • Input and output functions
 ഇൻപുട്ട്/ഔട്ട്പുട്ട് definition/meaning ഇൻപുട്ട്/ഔട്ട്പുട്ട്. ഇൻപുട്ട്/ഔട്ട്പുട്ട്
 ഇൻപുട്ട്/ഔട്ട്പുട്ട്, steps, application ഇൻപുട്ട്/ഔട്ട്പുട്ട് ഇൻപുട്ട്/ഔട്ട്പുട്ട്. Technical terms
 English-ൽ ഇൻപുട്ട്/ഔട്ട്പുട്ട് ഇൻപുട്ട്/ഔട്ട്പുട്ട്.

▼ • Control statements

• Control statements is an official topic in Module 1 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Control statements using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • control statements should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Control statements means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-നിയന്ത്രണ പ്രസ്താവനകൾ • Control statements നിയന്ത്രണ പ്രസ്താവനകൾ definition/meaning നിയന്ത്രണ പ്രസ്താവനകൾ. നിയന്ത്രണ പ്രസ്താവനകൾ, steps, application നിയന്ത്രണ പ്രസ്താവനകൾ നിയന്ത്രണ പ്രസ്താവനകൾ. Technical terms English-നിയന്ത്രണ പ്രസ്താവനകൾ.

▼ ◦ **if, if-else, nested if**

◦ if, if-else, nested if is an official topic in Module 1 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify ◦ if, if-else, nested if using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, ◦ if, if-else, nested if should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What ◦ if, if-else, nested if means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഓം if, if-else, nested if ഓം ഓം definition/meaning ഓം. ഓം ഓം, steps, application ഓം. Technical terms English-ഓം ഓം.

▼ ◦ switch

◦ switch is an official topic in Module 1 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify ◦ switch using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, ◦ switch should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What ◦ switch means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഓം switch ഓപ്പറേഷനുകൾക്ക് ഓരോന്നും definition/meaning ഓരോന്നും ഉണ്ടായിരിക്കണം. ഓരോന്നും ഓരോന്നും ഓരോന്നും, steps, application ഓരോന്നും ഓരോന്നും ഉണ്ടായിരിക്കണം. Technical terms English-ൽ ഉണ്ടായിരിക്കണം.

▼ ◦ while, do-while, for

◦ while, do-while, for is an official topic in Module 1 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify ◦ while, do-while, for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, ◦ while, do-while, for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What ◦ while, do-while, for means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഓ. while, do-while, for ഓരോ ഓരോ definition/meaning ഓരോ ഓരോ. ഓരോ ഓരോ steps, application ഓരോ ഓരോ. Technical terms English-ഓ ഓരോ ഓരോ.

▼ ◦ **break, continue**

◦ break, continue is an official topic in Module 1 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify ◦ break, continue using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, ◦ break, continue should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What ◦ break, continue means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പ്രാരംഭം • break, continue പ്രയോഗങ്ങൾ, definition/meaning പ്രയോഗങ്ങൾ. പ്രയോഗങ്ങൾ, steps, application പ്രയോഗങ്ങൾ. Technical terms English-ൽ പ്രയോഗങ്ങൾ.

▼ Static and Dynamic memory

Static and Dynamic memory is an official topic in Module 1 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Static and Dynamic memory using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, static and dynamic memory should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Static and Dynamic memory means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-സ്റ്റാറ്റിക് ആന്റ് ഡൈനാമിക് മെമറി
സ്റ്റാറ്റിക് മെമറി definition/meaning സ്റ്റാറ്റിക് മെമറി.
സ്റ്റാറ്റിക് മെമറി, steps, application സ്റ്റാറ്റിക് മെമറി സ്റ്റാറ്റിക് മെമറി. Technical terms
English-ൽ സ്റ്റാറ്റിക് മെമറി സ്റ്റാറ്റിക് മെമറി.

▼ Arrays

Arrays is an official topic in Module 1 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Arrays using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, arrays should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Arrays means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-**Arrays** **അറേ** **definition/meaning** **അറേയുടെ അർത്ഥം**. **അറേയുടെ** **അവസ്ഥ** **അറേയുടെ**, **steps**, **application** **അറേയുടെ** **അവസ്ഥ** **അറേയുടെ**. **Technical terms English-**അറേ**** **അറേയുടെ**.

▼ • Declaration & Initialization

• Declaration & Initialization is an official topic in Module 1 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Declaration & Initialization using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • declaration & initialization should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Declaration & Initialization means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- $\square\square$ • Declaration & Initialization

$\square\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square$, steps, application $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square\square$.

▼ • One-dimensional arrays

• One-dimensional arrays is an official topic in Module 1 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • One-dimensional arrays using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • one-dimensional arrays should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • One-dimensional arrays means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഏകദimensonal arrays ഏകദimensonal arrays definition/meaning ഏകദimensonal arrays. ഏകദimensonal arrays steps, application ഏകദimensonal arrays. Technical terms English-ഏകദimensonal arrays.

▼ • Array operations- Insert and Delete

• Array operations- Insert and Delete is an official topic in Module 1 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Array operations- Insert and Delete using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • array operations- insert and delete should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Array operations- Insert and Delete means in the official course context

MODULE 2

Arrays & Strings • Two-dimensional arrays

11 4

This module is presented from the official syllabus scope for Fundamentals of Computer Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Arrays & Strings • Two-dimensional arrays 11 4
- Official duration: As specified
- Official topic entries captured: 13
- The full source excerpt is preserved below for coverage checking.

▼ Arrays & Strings • Two-dimensional arrays 11 4

Arrays & Strings • Two-dimensional arrays 11 4 is an official topic in Module 2 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Arrays & Strings • Two-dimensional arrays 11 4 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, arrays & strings • two-dimensional arrays 11 4 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Arrays & Strings • Two-dimensional arrays 11 4 means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
Arrays & Strings • Two-dimensional arrays 11 4 definition/meaning .
steps, application . Technical terms English-
.

▼ Strings

Strings is an official topic in Module 2 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Strings using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, strings should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Strings means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-**Strings** **രണ്ടാം മോഡ്യൂളിലെ** **Strings** **എന്നതിന്റെ** **definition/meaning** **വിശദീകരിക്കുക.** **രണ്ടാം മോഡ്യൂളിലെ** **Strings** **എന്നതിന്റെ** **steps,** **application** **എന്നിവയെ** **വിശദീകരിക്കുക.** **Technical terms** **English-ൽ** **എന്നിവ** **എഴുതുക.**

▼ • String operations: length, compare, concatenate, copy

• String operations: length, compare, concatenate, copy is an official topic in Module 2 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • String operations: length, compare, concatenate, copy using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • string operations: length, compare, concatenate, copy should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • String operations: length, compare, concatenate, copy means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
 • String operations: length, compare, concatenate, copy
 definition/meaning.
 steps, application
 Technical terms English-

▼ • String matching

• String matching is an official topic in Module 2 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • String matching using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • string matching should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • String matching means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-**String matching** **definition/meaning** **steps**, application **Technical terms English-** **Malayalam**.

▼ • Built-in functions of string

• Built-in functions of string is an official topic in Module 2 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Built-in functions of string using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • built-in functions of string should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Built-in functions of string means in the official course context
Study lens	Principle → components or stages → result → application

▼ Structure and Union

Structure and Union is an official topic in Module 2 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Structure and Union using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, structure and union should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Structure and Union means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-**Structure and Union** **രൂപരേഖ** **രൂപരേഖ** definition/meaning **രൂപരേഖ**. **രൂപരേഖ** **രൂപരേഖ** **രൂപരേഖ**, steps, application **രൂപരേഖ** **രൂപരേഖ** **രൂപരേഖ**. Technical terms English-**രൂപരേഖ** **രൂപരേഖ**.

▼ Function

Function is an official topic in Module 2 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Function using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, function should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Function means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- $\square\square$ Function $\square\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square\square\square\square\square\square\square$, steps, application $\square\square\square\square\square\square\square\square\square\square$ $\square\square\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square\square\square\square\square\square\square$.

▼ Function prototype, definition, and call

Function prototype, definition, and call is an official topic in Module 2 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Function prototype, definition, and call using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, function prototype, definition, and call should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Function prototype, definition, and call means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
 Function prototype, definition, and call
 definition/meaning.
 , steps, application . Technical terms
 English- .

▼ • Add, subtract and multiply two numbers

• Add, subtract and multiply two numbers is an official topic in Module 2 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Add, subtract and multiply two numbers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • add, subtract and multiply two numbers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Add, subtract and multiply two numbers means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- $\square$ • Add, subtract and multiply two numbers $\square$ $\square$ definition/meaning $\square$. $\square$ $\square$ $\square$ $\square$, steps, application $\square$ $\square$ $\square$. Technical terms English- $\square$ $\square$ $\square$.

▼ • Factorial of a number

• Factorial of a number is an official topic in Module 2 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Factorial of a number using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • factorial of a number should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Factorial of a number means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഫാക്ടോറിയൽ • Factorial of a number ഫാക്ടോറിയൽ നമ്പറിന്റെ definition/meaning ഫാക്ടോറിയൽ നമ്പറിന്റെ. ഫാക്ടോറിയൽ നമ്പറിന്റെ steps, application ഫാക്ടോറിയൽ നമ്പറിന്റെ application. Technical terms English-ൽ ഫാക്ടോറിയൽ നമ്പറിന്റെ.

▼ Recursion

Recursion is an official topic in Module 2 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Recursion using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, recursion should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Recursion means in the official course context
Study lens	Principle → components or stages → result → application

Malayalam support: Module 2-Recursion പരസ്പരം വിളിക്കുന്ന പ്രക്രിയ definition/meaning പരിഭാഷിക്കുക. പ്രശ്നം പരിഹാരം പ്രശ്നങ്ങൾ, steps, application പ്രയോഗം പ്രയോഗങ്ങൾ പ്രയോഗങ്ങൾ. Technical terms English-പരിഭാഷിക്കുക പരിഹാരം.

▼ • Fibonacci series using recursion

• Fibonacci series using recursion is an official topic in Module 2 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Fibonacci series using recursion using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • fibonacci series using recursion should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Fibonacci series using recursion means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഫൈബോനാച്ചി സീരീസ് ഉപയോഗിച്ച് റിക്കർഷൻ
 ഫൈബോനാച്ചി സീരീസ് definition/meaning ഫൈബോനാച്ചി സീരീസ്. ഫൈബോനാച്ചി സീരീസ്
 ഫൈബോനാച്ചി, steps, application ഫൈബോനാച്ചി ഫൈബോനാച്ചി ഫൈബോനാച്ചി. Technical terms
 English-ഫൈബോനാച്ചി ഫൈബോനാച്ചി.

▼ • Binary search using recursion

• Binary search using recursion is an official topic in Module 2 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Binary search using recursion using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • binary search using recursion should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Binary search using recursion means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- $\square\square$ • Binary search using recursion
 $\square\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square$ $\square\square\square\square\square$
 $\square\square\square\square\square\square$, steps, application $\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square$. Technical terms
 English- $\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square\square\square$.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Searching and Sorting 12 4

This module is presented from the official syllabus scope for Fundamentals of Computer Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Searching and Sorting 12 4
- Official duration: As specified
- Official topic entries captured: 13
- The full source excerpt is preserved below for coverage checking.

▼ Searching and Sorting 12 4

Searching and Sorting 12 4 is an official topic in Module 3 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Searching and Sorting 12 4 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, searching and sorting 12 4 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Searching and Sorting 12 4 means in the official course context
Study lens	Principle → components or stages → result → application

Definition, labelled explanation, comparison, procedure or example as appropriate

000000000000 00000 definition/meaning 000000000000. 000000 000000
 00000000, steps, application 000000 0000000000 000000. Technical terms
 English-0 00000 000000000000.

▼ Searching techniques

Searching techniques is an official topic in Module 3 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Searching techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, searching techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Searching techniques means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
Searching techniques
 definition/meaning .
 , steps, application . Technical terms English-
 .

▼ • Linear search

• Linear search is an official topic in Module 3 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Linear search using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • linear search should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Linear search means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
• Linear search
definition/meaning
application
Technical terms English-
.

▼ • Binary search

• Binary search is an official topic in Module 3 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Binary search using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • binary search should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Binary search means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
 • Binary search
 definition/meaning
 .
 , steps,
 application
 . Technical terms English-
 .

▼ Sorting techniques

Sorting techniques is an official topic in Module 3 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Sorting techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, sorting techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Sorting techniques means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

▼ • Selection sort

• Selection sort is an official topic in Module 3 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Selection sort using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • selection sort should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Selection sort means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
 • Selection sort definition/meaning .
 . steps,
 application . Technical terms English-
 .

▼ • Insertion sort

• Insertion sort is an official topic in Module 3 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Insertion sort using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • insertion sort should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Insertion sort means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
 • Insertion sort
 definition/meaning
 application
 steps,
 application
 Technical terms English-
 .

▼ • Bubble sort

• Bubble sort is an official topic in Module 3 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Bubble sort using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • bubble sort should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Bubble sort means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പാഠ്യ • Bubble sort പദാർത്ഥം പദാർത്ഥം definition/meaning പദാർത്ഥം പദാർത്ഥം. പദാർത്ഥം പദാർത്ഥം പദാർത്ഥം, steps, application പദാർത്ഥം പദാർത്ഥം പദാർത്ഥം. Technical terms English-ൽ പദാർത്ഥം പദാർത്ഥം പദാർത്ഥം.

▼ Pointers

Pointers is an official topic in Module 3 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Pointers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, pointers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Pointers means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പുറം Pointers പദപരിചയം പദപരിചയം definition/meaning പദപരിചയം പദപരിചയം. പദപരിചയം പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-പദ പദപരിചയം പദപരിചയം.

▼ • Pointer operators

• Pointer operators is an official topic in Module 3 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Pointer operators using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • pointer operators should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Pointer operators means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പുറം • Pointer operators പരസ്പരം പരസ്പരം definition/meaning പരസ്പരം പരസ്പരം. പരസ്പരം പരസ്പരം പരസ്പരം, steps, application പരസ്പരം പരസ്പരം പരസ്പരം. Technical terms English-പുറം പരസ്പരം പരസ്പരം പരസ്പരം.

▼ • Parameter passing

• Parameter passing is an official topic in Module 3 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Parameter passing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • parameter passing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Parameter passing means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പാരാമീറ്റർ പാസ്സിംഗ് • Parameter passing പാരാമീറ്റർ പാസ്സിംഗ് definition/meaning പാരാമീറ്റർ പാസ്സിംഗ്. പാരാമീറ്റർ പാസ്സിംഗ് പാരാമീറ്റർ, steps, application പാരാമീറ്റർ പാസ്സിംഗ് പാരാമീറ്റർ. Technical terms English-പാരാമീറ്റർ പാസ്സിംഗ് പാരാമീറ്റർ.

▼ ◦ Pass by value

◦ Pass by value is an official topic in Module 3 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify ◦ Pass by value using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, ◦ pass by value should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What ◦ Pass by value means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ാം ഘട്ടം • Pass by value പരമ്പരപരമായ പരമ്പര definition/meaning പരമ്പരപരമായ പരമ്പര. പരമ്പര പരമ്പര പരമ്പര, steps, application പരമ്പര പരമ്പരപരമായ പരമ്പര. Technical terms English-ൽ പരമ്പര പരമ്പരപരമായ.

▼ ◦ Pass by reference

◦ Pass by reference is an official topic in Module 3 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify ◦ Pass by reference using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, ◦ pass by reference should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What ◦ Pass by reference means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പാഠ്യം. Pass by reference പരസ്പരം പരസ്പരം definition/meaning പരസ്പരം പരസ്പരം. പരസ്പരം പരസ്പരം പരസ്പരം, steps, application പരസ്പരം പരസ്പരം പരസ്പരം. Technical terms English-ൽ പരസ്പരം പരസ്പരം പരസ്പരം.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Function, Pointers & File 11 4

This module is presented from the official syllabus scope for Fundamentals of Computer Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Function, Pointers & File 11 4
- Official duration: As specified
- Official topic entries captured: 9
- The full source excerpt is preserved below for coverage checking.

▼ Function, Pointers & File 11 4

Function, Pointers & File 11 4 is an official topic in Module 4 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Function, Pointers & File 11 4 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, function, pointers & file 11 4 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Function, Pointers & File 11 4 means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- $\square\square$ Function, Pointers & File 11 4

$\square\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square$, steps, application $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square\square$.

▼ • File concepts

• File concepts is an official topic in Module 4 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • File concepts using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • file concepts should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • File concepts means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഫയൽ • File concepts ഫയൽ കോൺസെപ്റ്റ് definition/meaning ഫയൽ കോൺസെപ്റ്റ്. ഫയൽ കോൺസെപ്റ്റ്, steps, application ഫയൽ കോൺസെപ്റ്റ് ഫയൽ കോൺസെപ്റ്റ്. Technical terms English-ൽ ഫയൽ കോൺസെപ്റ്റ്.

▼ • Sequential file handling

• Sequential file handling is an official topic in Module 4 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Sequential file handling using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • sequential file handling should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Sequential file handling means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-**•** Sequential file handling **രണ്ടാം** **രണ്ടാം** definition/meaning **രണ്ടാം**. **രണ്ടാം** **രണ്ടാം** **രണ്ടാം**, steps, application **രണ്ടാം** **രണ്ടാം** **രണ്ടാം**. Technical terms English-**രണ്ടാം** **രണ്ടാം** **രണ്ടാം**.

▼ • Random access file handling

• Random access file handling is an official topic in Module 4 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Random access file handling using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • random access file handling should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Random access file handling means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഏകദേശം • Random access file handling

ഏകദേശം ഏകദേശം definition/meaning ഏകദേശം. ഏകദേശം ഏകദേശം ഏകദേശം, steps, application ഏകദേശം ഏകദേശം ഏകദേശം. Technical terms English-ഏകദേശം ഏകദേശം ഏകദേശം.

▼ • ● File operations using C - Read, Write and Append

• ● File operations using C - Read, Write and Append is an official topic in Module 4 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • ● File operations using C - Read, Write and Append using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • ● file operations using c - read, write and append should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • ● File operations using C - Read, Write and Append means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
 • ● File operations using C - Read, Write and Append
 definition/meaning.
 steps, application
 Technical terms English-

▼ Detailed Syllabus

Detailed Syllabus is an official topic in Module 4 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Detailed Syllabus using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, detailed syllabus should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Detailed Syllabus means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്ന Detailed Syllabus പരസ്യപ്പെടുത്തിയിരിക്കുന്നു. definition/meaning പരസ്യപ്പെടുത്തിയിരിക്കുന്നു. പരസ്യപ്പെടുത്തിയിരിക്കുന്നു, steps, application പരസ്യപ്പെടുത്തിയിരിക്കുന്നു പരസ്യപ്പെടുത്തിയിരിക്കുന്നു. Technical terms English-ൽ പരസ്യപ്പെടുത്തിയിരിക്കുന്നു.

▼ Mode of

Mode of is an official topic in Module 4 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mode of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mode of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mode of means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നിരിക്കുന്നതാണ് മോഡ്യൂൾ 4-ന്റെ നിർവ്വചനം/അർത്ഥം. നിർവ്വചനം, അർത്ഥം, ഏതെങ്കിലും ഉദാഹരണം, steps, application എന്നിവയെക്കുറിച്ച് വിശദീകരിക്കുക. Technical terms English-ൽ എഴുതുക.
 നിർവ്വചനം/അർത്ഥം.

▼ Instructional

Instructional is an official topic in Module 4 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Instructional using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, instructional should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Instructional means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഇന്റ്രക്ഷണൽ ഇൻസ്ട്രക്ഷണൽ ഡിഫിനിഷൻ/മീനിംഗ് ഇന്റ്രക്ഷണൽ ഇൻസ്ട്രക്ഷണൽ. ഇന്റ്രക്ഷണൽ ഇൻസ്ട്രക്ഷണൽ ഇൻസ്ട്രക്ഷണൽ, steps, application ഇന്റ്രക്ഷണൽ ഇൻസ്ട്രക്ഷണൽ ഇൻസ്ട്രക്ഷണൽ. Technical terms English-ഇന്റ്രക്ഷണൽ ഇൻസ്ട്രക്ഷണൽ.

▼ Subtopic Student Learning Outcome delivery Mapped CO Taxonomy Level

Subtopic Student Learning Outcome delivery Mapped CO Taxonomy Level is an official topic in Module 4 of Fundamentals of Computer Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Subtopic Student Learning Outcome delivery Mapped CO Taxonomy Level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, subtopic student learning outcome delivery mapped co taxonomy level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Subtopic Student Learning Outcome delivery Mapped CO Taxonomy Level means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Subtopic Student Learning Outcome delivery Mapped CO Taxonomy Level definition/meaning . steps, application . Technical terms English- .

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

- for Self-Learning
- Week Self-Learning description Hours Module I
- Understand different programming paradigms. Identify the structure of a C program
- Explain the applications of C language.
- Use data types, variables. 5 expressions, and identify the structure Module II
- Understand static and dynamic memory
- Work with one-dimensional and two-dimensional array
- Perform array operations like insertion and deletion
- Apply string operations and built-in string functions Module III
- Identify prototype, definition, and call in a given program. 5
- Write a program using a user-defined function.
- Use built-in functions from standard libraries. 5
- Identify appropriate header files for built-in functions.
- Implement string length, copy, and compare without using library functions.
- Compare two strings and display whether they are equal. 5
- Compare recursive and iterative solutions.
- Demonstrate pointer initialization and dereferencing
- Modify variable value using pointer.
- Explain pointer arithmetic with array examples.
- Traverse an array using pointers.
- Pass array to a function using pointers.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

– Theory						
		CA1		CA2		
CA3	CA4	CA5	CA6		ESE	
Mode	Attendance	Self-learning	Self-learning			Self-
learning	Self-learning	Series	Series		Written	
		activities			activities	
activities	activities	Exam	Exam			Exam
		(Module 1)	(Module 2)			
(Module 3)	(Module 4)	(2 modules)	(2 modules)			(theory)
Duration						
2 hours	2 hours	2.5 hours				
Exam mark		15		15		
15	15	50	50		60	
Converted to						
20	20					
Marks	5				15	
20	60					
Diploma Curriculum Revision 2026						2432
Page #4						
Total						
40						60
Tentative	End of					
End of						

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

- ▶ **Define or explain C Fundamentals 11 3. (3 marks)**
- ▶ **Define or explain • Input and output functions. (3 marks)**
- ▶ **Define or explain • Control statements. (3 marks)**
- ▶ **Define or explain • if, if-else, nested if. (3 marks)**

Module 2

- ▶ **Define or explain Arrays & Strings • Two-dimensional arrays 11 4. (3 marks)**
- ▶ **Define or explain Strings. (3 marks)**
- ▶ **Define or explain • String operations: length, compare, concatenate, copy. (3 marks)**
- ▶ **Define or explain • String matching. (3 marks)**

Module 3

- ▶ **Define or explain Searching and Sorting 12 4. (3 marks)**

► **Define or explain Searching techniques. (3 marks)**

► **Define or explain • Linear search. (3 marks)**

► **Define or explain • Binary search. (3 marks)**

Module 4

► **Define or explain Function, Pointers & File 11 4. (3 marks)**

► **Define or explain • File concepts. (3 marks)**

► **Define or explain • Sequential file handling. (3 marks)**

► **Define or explain • Random access file handling. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **C Fundamentals 11 3**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Arrays & Strings • Two-dimensional arrays 11 4**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Searching and Sorting 12 4**.

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **Function, Pointers & File 11 4**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- URL Modules Covered

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 2432. Verify institutional updates against the current official curriculum.